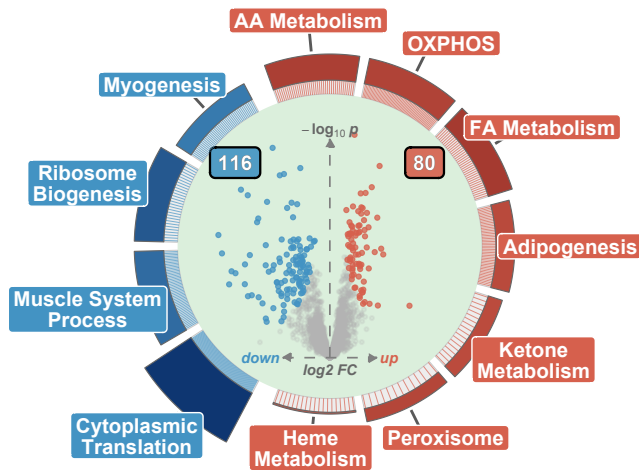


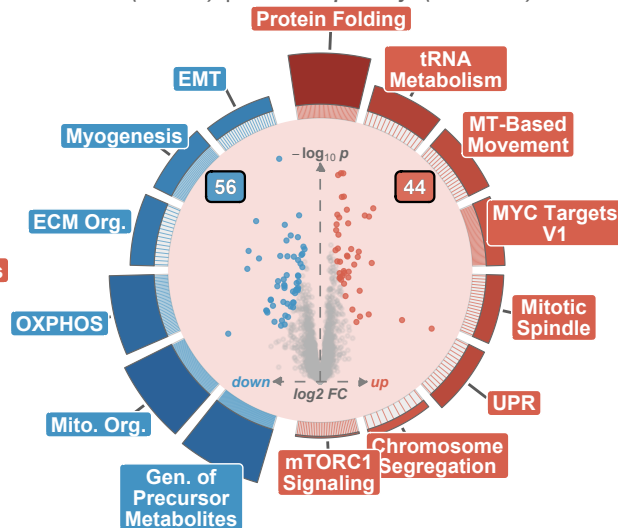
A Aging Effect

196 DEPs ($\pi < 0.05$) | 221 / 1985 pathways (FDR < 0.05)



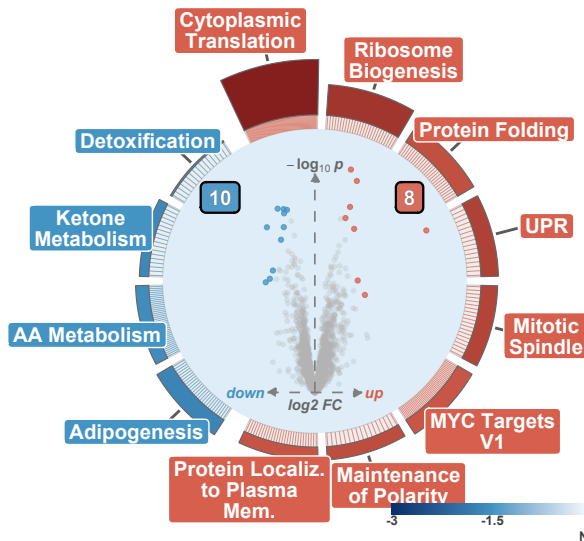
B Training Response (Young)

100 DEPs ($\pi < 0.05$) | 313 / 1982 pathways (FDR < 0.05)



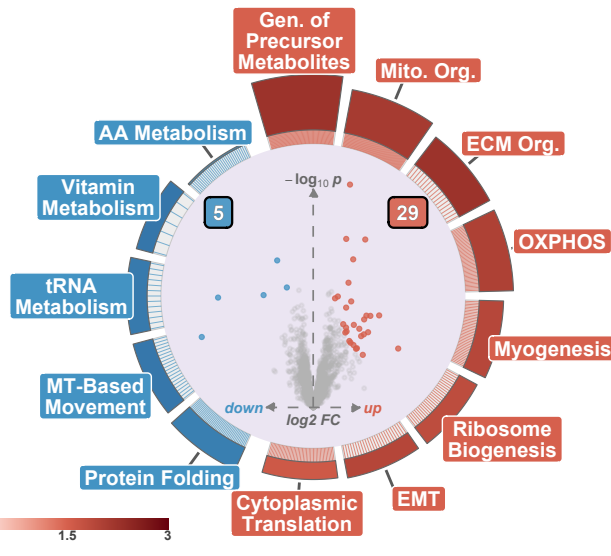
C Training Response (Old)

18 DEPs ($\pi < 0.05$) | 84 / 1999 pathways (FDR < 0.05)



D Age × Training Interaction

34 DEPs ($\pi < 0.05$) | 277 / 1982 pathways (FDR < 0.05)



Point colour marks the $\pi < 0.05$ frontier ($\pi = P \times |\log_2 FC|$), a curved magnitude–significance boundary; volcano axes are per-contrast relative, not absolute. Ring arc height encodes term rank, not padj. Panel A is a cross-sectional baseline age difference (Old_Pre – Young_Pre); B–D are longitudinal training and interaction effects.